Hrushikesh Etikikota

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Educational Qualification

Degree	Institution	CPI/%	Year
M.Tech	IIT Gandhinagar	8.71	2022 - 2024
B. Tech (CSE)	Sri Venkateswara University	8.68	2017 - 2021
Class XII	Sri Siddhartha Junior College (BIEAP)	98.8%	2015 – 2017
Class X	SPVB High School (BSEAP)	9.8	2014 - 2015

Experience

LLM – Python Engineer, Turing Enterprises Inc. **[Oct 2024 - Present]**

- Created over 300 notebooks for fine-tuning of LLMs upon testing across the projects: CoT Reasoning, Function Calling (FC), Agent Completion, Adversarial Self-Correction, Vision FC, and Structured output.
- Promoted to Team Lead for a team of 10 engineers across Agent Completion and Vision-FC projects while consistently doing review of 20 and 50 tasks per day respectively to meet project requirements.
- Assessed the quality and safety of 100+ bilingual conversations (Hindi/English) for an RLHF project.
- Developed Python-based test suites on RLHF project for public GitHub repositories where the LLM models failed to generate correct test cases and evaluate their ability to add new features/ functionalities as well.

Teaching Assistant at IIT Gandhinagar **[July 2022 - Jun 2024]**

- Delivered Python lab support and managed exam logistics for 450+ students alongside a 20-member team.
- Assisted 160+ students with assignments, providing guidance on coursework in Databases course.
- Guided 20+ students in Verilog labs, receiving recognition from the professor for exceptional performance.

Freelance Tutor on Course Hero Platform **[Jun 2021 - Jul 2022]**

- Supported over 250 students on their assignments and homework with a 96% helpful rate that enhanced my problem-solving skills on various computer science topics by engaging with a diverse student community.

Projects

A Gaze-Controlled Typing Interface with Context-Based Letter and Word Predictions

- Designed a multi-level tree-based virtual keyboard that supports severe motor and speech-impaired people.
- Integrated statistical letter prediction and word prediction models that help with quick and easy typing.
- Conducted experiments with Mouse and Eye-Tracker and compared the results with the Dasher software
- Obtained a typing speed of 30.70 and 18.47 letters per minute using the mouse and the eye tracker, respectively, which are nearly 2.7 times more than that of the Dasher with both input devices.

Helmet Detection and License Plate Extraction Using Yolov3 and Darknet

- Trained a real-time object detection algorithm - YOLOv3 model on 90% of the dataset of 4000 images using the Darknet Framework, achieving an impressive 87.49% average precision on a 10% test set.
- Created a new labelled dataset of 2800 images using LabelImg software and trained the model on 90% of it to achieve a remarkable 97.51% average precision the 10% test set.

ML Projects: Airfoil self-noise Prediction, Breast Cancer Detection, Loan Prediction, etc.

- Executed ML projects by implementing the core algorithms from scratch and evaluated their performance.
- Utilized appropriate models based on the project, resulting in valuable insights and practical experience.

Transformer Architectures for Natural Language Processing

- Engineered 6-layer Decoder and 4-layer Encoder-Decoder Transformers from the ground up in PyTorch to understand the internal architectural mechanics through NMT and next-character prediction projects.

Paper Publications

- Published the paper titled "Predictive Tree-based Virtual Keyboard for Improved Gaze Typing" on IEEE SMC2024.

Online Course Certifications

- ‘**Getting Started with AWS Machine Learning**’ by AWS Specialization.
- ‘**Data Science Math Skills**’ by Duke University.
- ‘**THE FUNDAMENTALS OF DIGITAL MARKETING certification’** exam by GOOGLE DIGITAL UNLOCKED.
- **‘Python for Everybody’** by the University of Michigan.

Achievements

- Received PRATIBHA AWARD for Academic Excellence in the 10th standard issued by Govt. of AP.
- Secured 98.41 percentile (2,981 rank in 187484 students) in EAMCET 2017 (Govt. of AP).
- Attained 98.74 percentile (974 rank in 77257 students) in Graduate Aptitude Test in Engineering (GATE 2022).
- Solved 500+ questions and earned badges on platforms like Hacker Rank, Coding Ninjas, and LeetCode.

Skill Summary

- Core Software & Systems: Python, C, C++, SQL, HTML/CSS, Django, Git, DSA, OOPs, Databases, OS
- AI & Machine Learning: GenAI, LLM-assisted Development, Prompt Engineering, PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, YOLO, Transformer Architecture